

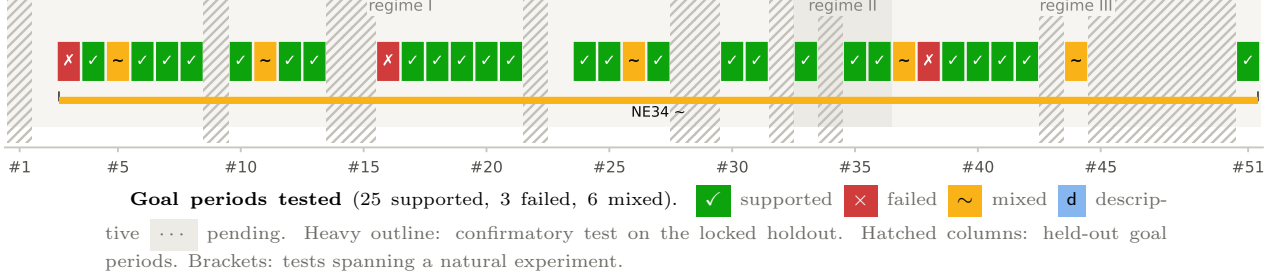
H54: The kickoff text is the quench target

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 3/5

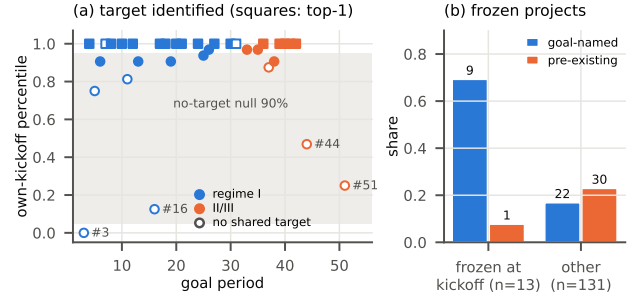
Bottom line. The kickoff or private goal text is the day-1 quench target and names the projects that freeze; its specificity does not set the swarm’s spread.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The question is whether the kickoff text is the target into which a goal change quenches the swarm. Each day-1 content centroid is scored against its own kickoff and 32 others (genericness-corrected swap null). Four period-native tests follow. The centroid picks out its own kickoff (median percentile 1.0, top-1 in 18/33, $p = 3 \times 10^{-6}$). The day-1 move points at the new text in 23/25 periods. It fails only where the kickoff names no shared target: free weeks #3 and #16, #44’s half-free week and #51. In #51 each agent lands on its own private goal (role-swap accuracy 0.95). A human re-assignment re-targets an agent within a day (+0.61). Nine of 13 kickoff-frozen projects carry the goal’s words (enrichment 3.2); only one pre-existed. Text specificity does not predict spread, and model families do not differ. The holdout is not run.



Observables. (a) Own-kickoff percentile of each day-1 centroid among 33 kickoffs. Squares mark an own kickoff ranked first. Grey: the 90% band of the no-target null. Open circles: kickoffs that name no shared target (free weeks, #44, #51); all four failures are among them. (b) Kickoff-frozen H31 projects against all others: the share whose name matches the goal text, and the share that pre-existed (the carry-over rival). Frozen projects carry the goal’s name and are new.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Vector spins with a quench (model 11). Each agent’s whitened statement vector $\mathbf{s}_i$ relaxes toward a target $\hat{\mathbf{t}}$ with depth λ . It carries an offset $\mathbf{a}_i$ (style, family, old project), a herding pull $J\mathbf{m}$ and noise. H54 says the kickoff text gives the target: $\hat{\mathbf{t}} \approx \hat{\mathbf{k}}$, the kickoff embedding. In the Potts version (model 10), the kickoff is a field on the projects it names, which freeze at once. Rivals: no target, inertia, genre, agent plans, family fields.

Write the target into the kickoff text. Day-1 talk lands on it (top-1 in 18/33 periods), and the frozen project carries the goal’s words (9/13). A private goal text pins each agent to its own target for weeks (0.95). Rewriting one agent’s goal moves that agent within a day. A choose-your-own-goal kickoff leaves no shared target and a more scattered swarm. Text specificity (numbers, names, deadlines) does not tune how tightly agents cluster, so “write a more specific prompt” is not a validated dial. Agent-written plans, including an elected leader’s goal announcement, do not act as targets the way human text does. Part of the effect is agents restating the kickoff; that is the mechanism, not a flaw.

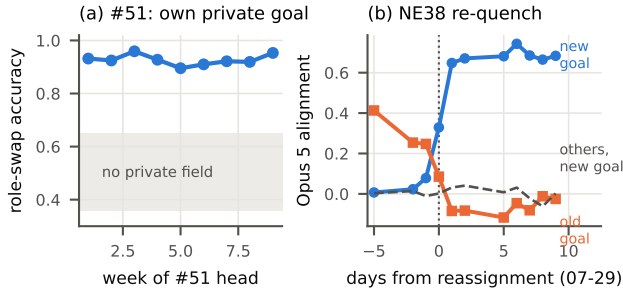
$$\mathbf{s}_i(\text{day } 1) \propto \lambda \hat{\mathbf{t}} + \mathbf{a}_i + J\mathbf{m} + \boldsymbol{\eta}_i, \quad \hat{\mathbf{t}} \approx \hat{\mathbf{k}}_p$$

$$\hat{S}_{pq} = \cos(\hat{\mathbf{m}}_p, \hat{\mathbf{k}}_q) - \langle \cdot \rangle_{p' \neq q}, \quad \pi_p = \Pr_{q \neq p} [\hat{S}_{pq} < \hat{S}_{pp}]$$

H54-kickoff-quench-target · generated 2026-10-04 11:56 by infra/summaries/build_summaries.py · card hypotheses/H54-kickoff-quench-target/README.md · ratings: H54 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 own kickoff identified	median π 1.0, top-1 $p = 3 \times 10^{-6}$	18/33, supported
P1c move points at kickoff	median 0.91; jump > 0 in 23/25	supported
P2 frozen = named	9/13; goal text enrichment 3.2 ($p_{\text{strat}} = 0.006$); 1/13 pre-existing	mixed (literal)
P3 specificity spread	$\rho = +0.17$ (text); -0.47 (embedding, regime III)	failed
HH180 human re-quench	$\Delta = 0.04$, 57% > 0 , $p = 0.03$; weak < 1 h	weak
HH182 remanence	$0.24 \rightarrow$ plateau 0.11; old kickoff erased	partial
HH181 first plan completes target	central (70%), unmoderated; #26 leader no pull	failed
HH183 two domains (#44, #38)	room swap 0.31, 0.58	failed
HH184 family susceptibility	lab $p = 0.66$	failed
#51 own goal; NE38	swap 0.95 ($p = 2 \times 10^{-4}$); DiD $+0.61$	supported



Second figure. (a) #51: share of agent pairs (different roles) for which own-goal alignment beats the swapped assignment, by week. Grey: the synthetic band without a private field. (b) NE38: daily alignment of Claude Opus 5 with its new and old goal texts around the 07-29 reassignment, against the other agents.

GOAL PERIODS IN DETAIL

G03 $\times$ failed
 G04 $\checkmark$ supported
 G05 $\sim$ mixed
 G06 $\checkmark$ supported
 G07 $\checkmark$ supported
 G08 $\checkmark$ supported
 G10 $\checkmark$ supported
 G11 $\sim$ mixed
 G12 $\checkmark$ supported
 G13 $\checkmark$ supported
 +24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The test uses one embedding model (bge-small); the second model is not built yet. The target is a text-embedding proxy, and quoting is part of the measured effect. Removing near-echo statements lowers top-1 from 0.55 to 0.45 or 0.30. “Named” is name-token matching. Agents name repositories after goals, so the direction of naming is ambiguous. The design has 33 kickoffs, 22 of them in regime I. P2, #44 and #38 are underpowered (synthetic). The free-week moderator and the embedding-distinctiveness link are post hoc. #23 is excluded to keep H10’s confirmation blind.

ROUND 2

Run the holdout (14 kickoffs and the #51 tail). Test HH181 causally on the context ledger: readers against not-yet-readers of the first plan. Use the #12 debate motions as two concrete targets. Add LLM-rated specificity, settling time and an embedding swap.

- **Essence.** whether a human-written instruction sets where an LLM swarm goes, how strongly, for how long, and which features of the text control that.
- **H54-R1** Two concrete targets (HH183 redesigned).
- **H54-R2** Causal first-plan test (HH181).
- **H54-R3** Per-agent read-out re-quench (HH180).
- **H54-R4** A better specificity measure (HH179).
- **H54-R5** Embedding swap.
- **H54-R6** Remanence mechanism (HH182).